

Subject Description Form

Subject Code	ME591
Subject Title	Dissertation
Credit Value	9
Level	5
Pre-requisite / Co-requisite/ Exclusion	Consent of Project Supervisor and endorsement of Dissertation Coordinator should be obtained before subject registration.
Objectives	To develop analytical and research skills to design and implement practitioner research in mechanical engineering or related fields.
Intended Learning Outcomes	Upon completion of the dissertation, students will be able to: a. demonstrate an understanding of relevant literature in the topic area selected; b. develop critical thinking and analytic evaluation skills through planning and implementing a research project; c. demonstrate the ability to enhance and apply advanced knowledge to solve complex engineering problems; d. communicate effectively the dissertation process, results, experience and reflection coherently, logically and precisely, using written, oral, and visual media.
Subject Synopsis/ Indicative Syllabus	The student may select, plan and conduct a research project in mechanical engineering or related field, subject to the availability of supervisor(s) and their research interests and background. The research area should be in line with the student's overall design of his/her chosen programme of study and choice of subjects.
Teaching/Learning Methodology	A topic for research will be mutually agreed upon between the student and project supervisor(s). The student will read widely on the scientific issues and, in specific areas, also in depth under the guidance of the supervisor(s). The student shall contact his/her project supervisor(s) from time to time to discuss the research progress and observations. The project supervisor(s) will provide guidance to the student and advice on the style of dissertation presentation, etc. The project supervisor(s) will be available for consultation at the University/at the student's workplace according to circumstances.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	1. Progress	20%	✓	✓	✓	✓
	2. Oral Exam	30%	✓	✓	✓	✓
	3. Dissertation	50%	✓	✓	✓	✓
	Total	100 %				
Student Study Effort Required	Class contact:					
	▪ Meeting and Oral Presentation			20 Hrs.		
	Other student study effort:					
	▪ Project/Research work			300 Hrs.		
	▪ Dissertation writing			100 Hrs.		
	Total student study effort			420 Hrs.		
Reading List and References	1. Grote, K. H., & Hefazi, H. (Eds.). (2021). <i>Springer handbook of mechanical engineering</i> . Springer Nature.					
	2. Davim, J. P. (Ed.). (2018). <i>Introduction to mechanical engineering</i> . Springer.					
	3. Stewart, R. (2022). <i>How to do research: And how to be a researcher</i> . Oxford University Press.					
	4. Calabrese, R. L. (2012). <i>Getting it right: The essential elements of a dissertation</i> . Rowman & Littlefield.					
	5. Randolph, J. (2019). <i>A guide to writing the dissertation literature review</i> . Practical assessment, research, and evaluation, 14(1), 13.					
Develop Date	January 2024					